

# Writing Logbook — Assignment 04

CSP529 Technical Writing & Publishing

*From Claim to Evidence: Can You Trust Your Source?*

MT26MCS024 — Gaurav Acharya    MT26MCS032 — Sandeep

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## 1. Assignment Overview

| Item               | Detail                                                       |
|--------------------|--------------------------------------------------------------|
| Practical Title    | From Claim to Evidence: Can You Trust Your Source?           |
| Stages Covered     | 9 stages — from statement identification through peer review |
| Group Members      | Gaurav Acharya (MT26MCS024), Sandeep (MT26MCS032)            |
| Sources Used       | AWS Blog, Chiari (2025), Desai et al. (2013, ACM SIGPLAN)    |
| Date of Completion | September 2026                                               |

## 2. Objectives & Learning Outcomes

This was the most comprehensive assignment in the TWP course, spanning nine stages from identifying claims through finding, evaluating, and tracing evidence, to peer review. The learning outcomes included:

- Distinguishing between facts, claims, and inferences that require different levels of evidence.
- Finding appropriate evidence for each statement type.
- Evaluating source quality on two axes: credibility (C1) and relevance/support (C2).
- Building a claim-evidence trace — mapping a claim back to its underlying data and methodology.
- Identifying the “citation trap” — sources that are topically relevant but do not actually support the specific claim.
- Calibrating claim strength to match the available evidence.
- Verifying AI-suggested references for existence and accuracy (combating hallucination).
- Conducting peer review of citation-claim traceability.

## 3. Work Performed — Stage by Stage

### 3.1. Stage 1: Statements and Required Evidence

We selected three statements from our Assignment 03 article summary, each representing a different statement type:

1. **Established Fact:** “The AWS FIS managed extension is implemented as an AWS Lambda layer.”  
→ *Evidence needed:* Official AWS technical documentation or architecture schematics.
2. **Claim:** “The primary barriers for formal methods tools are steep learning curve, required specialized domain expertise and lack of user-friendly interfaces.”  
→ *Evidence needed:* Empirical studies on developer experience, survey data, or peer-reviewed usability analyses.

3. **Inference:** “The P Language sought a balance between mathematical thinking capabilities while being easier to understand by using state-machine-based algorithms...”  
 → *Evidence needed:* The original foundational research paper by P’s creators, or empirical developer usability studies.

### 3.2. Stage 2: Finding the Evidence

For each statement, we identified the strongest available source:

| Statement   | Source Selected                                        | Type             | Why Suitable                              |
|-------------|--------------------------------------------------------|------------------|-------------------------------------------|
| Statement 1 | AWS Management Blog                                    | Technical Blog   | Primary source from the developers        |
| Statement 2 | Chiari (2025), <i>J. Software: Evol. &amp; Process</i> | Research Article | Peer-reviewed empirical usability study   |
| Statement 3 | Desai et al. (2013), ACM SIGPLAN                       | Research Paper   | Original foundational paper by P creators |

### 3.3. Stage 3: Source Quality — Credibility & Relevance

We evaluated each source on two criteria:

- **Source 1 (AWS Blog):**
  - C1 (Credibility): Highly credible — published by AWS, primary authority on their own systems.
  - C2 (Relevance): Directly supports — explicitly details the Lambda layer implementation.
- **Source 2 (Chiari 2025):**
  - C1: Highly credible — recent (2025) peer-reviewed journal article with empirical methodology.
  - C2: Directly supports — analyses developer usability challenges and adoption barriers.
- **Source 3 (Desai et al. 2013):**
  - C1: Highly credible — peer-reviewed, published by P’s original creators in ACM.
  - C2: Directly supports — establishes the design philosophy and state-machine architecture.

### 3.4. Stage 4: Claim-Evidence Trace

For Statement 2 (formal methods barriers), we built a complete traceability chain:

1. **Claim:** “The primary barriers... are steep learning curve...”  
 ↓ **Source:** Chiari (2025).  
 ↓ **Specific finding:** Developer adoption hindered by UI complexity and mathematical domain knowledge requirements.  
 ↓ **Underlying evidence:** Survey data from software engineers.  
 ↓ **Methodology:** Empirical research using surveys/interviews.

**Verdict:** Yes, a reader can trace from the sentence back to the underlying survey data to verify the claim.

### 3.5. Stage 5: The Citation Trap

For each statement, we examined two alternative sources that appeared relevant but did not actually support the specific claim:

| Stmnt   | Source Examined               | Classification        | Why It Fails                                         |
|---------|-------------------------------|-----------------------|------------------------------------------------------|
| Stmnt 1 | dev.to AWS Heroes Blog        | Background only       | Community tutorial, no architectural proof           |
| Stmnt 1 | ResearchGate resilience paper | Background only       | Broad topic, lacks specific Lambda layer detail      |
| Stmnt 2 | AdaCore Blog                  | Background only       | Product-focused, no empirical survey data            |
| Stmnt 2 | Kneuper (1997)                | Contradicts/qualifies | Discusses theoretical limits, not modern UI barriers |
| Stmnt 3 | Microsoft Research Blog       | Partially supports    | High-level overview, lacks academic depth            |
| Stmnt 3 | GitHub p-org/P                | Partially supports    | Practical docs, no design philosophy discussion      |

**Key Learning:** A source can be credible, topically relevant, and still not support the specific claim being made. This is the “citation trap.”

### 3.6. Stage 6: Calibrating Claim Strength

For Statement 2, we wrote three versions with increasing strength:

1. **Weak (Possibility):** “...*may sometimes present* barriers...”
2. **Moderate (Observation):** “Developers *frequently report* that...”
3. **Strong (Established Fact):** “The *primary* barriers *are*...”

**Justified version:** Moderate (Version 2) — the Chiari study provides robust survey data showing widespread developer struggles, but declaring them universally as “the primary” barriers overstates certainty without a broader meta-analysis.

**Calibrated claim:** “Empirical studies indicate that developers frequently cite steep learning curves, the necessity for specialized domain expertise, and a lack of user-friendly interfaces as significant barriers to adopting formal verification tools.”

### 3.7. Stage 7: Proper Citation in $\LaTeX$

We built a `references.bib` database with nine entries across three source categories. Key observation:

**What  $\LaTeX$  did automatically:** When adding, removing, or reordering citations, BibTeX/BibLaTeX automatically renumbered all in-text citations and regenerated the reference list. For a thesis with 100–200 references, this automation prevents hours of manual error-prone renumbering.

### 3.8. Stage 8: AI Reference Challenge

We prompted an AI to provide three scholarly references for Statement 2 and verified each:

| AI Reference                | Exists? | Correct? | Relevant? | Supports? | Verdict                |
|-----------------------------|---------|----------|-----------|-----------|------------------------|
| Woodcock et al. (2009)      | Yes     | Yes      | Yes       | Partially | Reject (outdated)      |
| Smith & Jones (2023)        | No      | No       | N/A       | N/A       | Reject (Hallucination) |
| Gleirscher & Koschel (2019) | Yes     | Yes      | Yes       | Yes       | Use                    |

**Verification process:** Searched exact titles and DOIs in Google Scholar and ACM Digital Library; cross-referenced authors and publication years; read abstracts and conclusions of existing papers to confirm actual support for our claim.

**Critical finding:** 1 out of 3 AI-suggested references was a complete hallucination — fabricated title, fabricated authors. This demonstrates why AI-generated references must *never* be trusted without independent verification.

### 3.9. Stage 9: Peer Review

We exchanged PDFs with a peer group and evaluated citation traceability:

- **Claim reviewed:** “The AWS FIS managed extension is implemented as an AWS Lambda layer.”
- **Reviewer question:** If the citation number is removed, can a reader reconstruct why this source was selected?
- **Reviewer verdict:** Supported — the claim is highly specific to AWS infrastructure, naturally pointing to official AWS documentation.
- **Our response:** Agreed; specificity of the claim ensures high traceability to primary corporate sources.

## 4. Individual Reflections

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1. **Credible vs. supportive:** A credible source is trustworthy and authoritative; a supportive source contains the specific data/fact needed to prove the exact statement. A highly credible paper on formal methods theory does *not* support a claim about user-interface barriers.
2. **Rejected sources:** We rejected the AdaCore blog — credible company, but the blog lacked the empirical survey data needed to prove industry-wide adoption barriers.
3. **Evidence changed wording:** Examining the evidence forced us to change from absolute (“The primary barriers *are*”) to calibrated (“Empirical studies *indicate*... developers *frequently cite*...”).
4. **Future citation practice:** Before citing any source, we will check if the specific sentence can be mapped to a specific table, graph, or concluding paragraph in the cited paper — true traceability, not title-based relevance.

## 5. Challenges Encountered

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1. **Finding the “right” source type:** Statement 1 was best supported by a corporate blog (primary source) rather than a peer-reviewed paper — contradicting the assumption that “peer-reviewed = always best.”
2. **Distinguishing “relevant” from “supportive”:** The citation trap exercise was the most conceptually difficult — many sources felt relevant but failed the support test.
3. **Calibrating language:** Finding the right hedging words (“indicate,” “frequently cite”) to match the evidence strength required multiple iterations.
4. **AI hallucination detection:** The fabricated Smith & Jones reference had realistic-sounding metadata, making it initially difficult to identify as fake.
5. **~~LaTeX~~ longtable formatting:** Multi-page tables with proper headers and footers required learning the `longtable` package’s `\endfirsthead`, `\endhead`, `\endfoot`, `\endlastfoot` system.

## 6. Key Takeaways

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1. The chain from claim  $\rightarrow$  source  $\rightarrow$  finding  $\rightarrow$  data  $\rightarrow$  methodology must be traceable end-to-end.
2. Source credibility and source support are *independent* dimensions — a source can be highly credible without supporting a specific claim.
3. Claim strength must be calibrated to evidence strength — overstating certainty is an integrity failure.
4. AI-generated references are unreliable and must be independently verified in academic databases before use.
5.  $\LaTeX$  reference management (BibTeX/BibLaTeX) automates numbering and formatting, making it indispensable for large documents.
6. Peer review of citation traceability reveals weaknesses that self-review misses.